Artificial Intelligence-Based Early Diabetes Prediction Using Machine Learning

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Abstract

Early identification of diabetes risk can support timely clinical assessment and preventive intervention. This study investigates machine-learning approaches for early diabetes prediction using the Pima Indians Diabetes Dataset, which contains 768 records and eight predictive attributes: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age. Physiologically implausible zero measurements in selected variables were treated as missing values and median-imputed within model pipelines. Five classification algorithms—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbours—were evaluated. The methodology combines a stratified 80:20 train-test split with five-fold stratified cross-validation on the training portion. Random Forest hyperparameters were tuned using randomized search with F1-score as the optimization criterion. In addition to accuracy, precision, recall, F1-score, and ROC-AUC were reported. Permutation importance was used to examine the contribution of input variables, while confusion-matrix and learning-curve analyses were used for model diagnostics. Under the primary held-out test split, the tuned Random Forest achieved 74.03% accuracy and 81.54% ROC-AUC. The results demonstrate that model selection should consider multiple evaluation measures rather than accuracy alone. The study is intended as a machine-learning research evaluation and not as a substitute for professional medical diagnosis.

Keywords: Artificial Intelligence, Machine Learning, Diabetes Prediction, Pima Indians Diabetes Dataset, Random Forest, Support Vector Machine

1. Introduction

Diabetes mellitus is a chronic metabolic disease marked by high levels of glucose in the blood and a deficiency either in insulin action or in insulin secretion. Since the disease may advance without showing many symptoms in its early phase, it is a key aim of both research and public health to identify those people who may be at increased risk. Instead of depending on clinical tests and professional judgement, machine-learning techniques are able to examine combinations of demographic,

physiological, and laboratory-related features in order to estimate risk patterns. These models can therefore assist research aimed at screening by highlighting patterns that deserve further investigation.

This study focuses on the Pima Indians Diabetes Dataset and compares several commonly used supervised-learning algorithms. The work extends a basic SVM implementation by adding a reproducible multi-model evaluation framework, five-fold cross-validation, Random Forest hyperparameter tuning, feature-importance analysis, and diagnostic visualizations. The objective is not to claim clinical deployment, but to evaluate how different classification approaches behave under a consistent experimental protocol.

2. Problem Statement

A simple single-model, single-split experiment can provide an incomplete view of diabetes-prediction performance. Results may vary with the selected train-test partition, and accuracy alone may hide differences in false-positive and false-negative behaviour. Therefore, a systematic comparison is required using multiple evaluation metrics, cross-validation, hyperparameter tuning, and diagnostic analysis while avoiding preprocessing leakage.

3. Objectives

- To preprocess the Pima diabetes dataset using a leakage-aware pipeline.
- To compare Logistic Regression, Decision Tree, Random Forest, SVM, and KNN.
- To combine a stratified 80:20 held-out test set with five-fold cross-validation.
- To tune Random Forest hyperparameters using randomized search and F1-score.
- To analyse feature contribution using permutation importance.
- To examine model behaviour using confusion matrices and a learning curve.

4. Dataset

The supplied diabetes.csv contains 768 observations and nine columns. Eight columns are predictors and Outcome is the binary target. The predictor variables are Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age. Outcome values represent the two target classes. The supplied CSV contains no literal missing entries; however, zero values in Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI were treated as missing measurements because zero is not physiologically plausible for these variables.

5. Methodology

5.1 Preprocessing

The data were divided into training and held-out test portions using an 80:20 stratified split with random_state=42. For variables containing implausible zeros, zero values were replaced with NaN. Median imputation and, where required, standardization were placed inside scikit-learn pipelines. This ensures that imputation and scaling parameters are learned from the relevant training data rather than from the complete dataset.

5.2 Models and Evaluation

Five classifiers were evaluated: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine with a probability-enabled classifier, and K-Nearest Neighbours. Five-fold Stratified K Fold cross-validation was applied to the training portion. Accuracy, precision, recall, F1-score, and ROC-AUC were used because the dataset contains unequal class frequencies and because diabetes screening performance cannot be characterized adequately by accuracy alone.

5.3 Random Forest Hyperparameter Tuning

Random Forest was tuned using Randomized Search CV with 10 sampled configurations and five-fold stratified cross-validation. F1-score was selected as the search objective. The tested parameter ranges included number of trees (100, 200, 300, 500), maximum depth (None, 5, 10, 15, 20), minimum samples for splitting (2, 5, 10), minimum samples per leaf (1, 2, 4), maximum features (sqrt, log2), and splitting criterion (gini, entropy). The selected configuration was:

n_estimators=300, max_depth=None, min_samples_split=5, min_samples_leaf=2, max_features=log2, criterion=gini.

5.4 Feature Importance and Diagnostics

Permutation importance was calculated on the held-out test set using F1-score and 20 random permutations per feature. This analysis is used as an explanatory diagnostic rather than as evidence of causal influence. Confusion matrices were generated for the tuned Random Forest, and a learning curve was generated for the Decision Tree using five-fold cross-validation.

6. Results

6.1 Five-Fold Cross-Validation Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	78.82 ± 1.91	76.38 ± 5.98	57.48 ± 2.27	65.45 ± 2.44	84.34 ± 1.85
Decision Tree	67.26 ± 2.34	53.06 ± 3.74	57.95 ± 5.02	55.19 ± 2.68	65.10 ± 2.11
Random Forest	76.06 ± 1.88	68.99 ± 4.23	57.50 ± 3.78	62.58 ± 2.74	81.63 ± 2.20
SVM	78.01 ± 1.36	74.15 ± 6.44	58.42 ± 5.72	64.85 ± 1.87	83.33 ± 2.28
KNN	72.96 ± 4.07	63.67 ± 8.94	56.54 ± 5.58	59.39 ± 4.69	78.43 ± 3.73

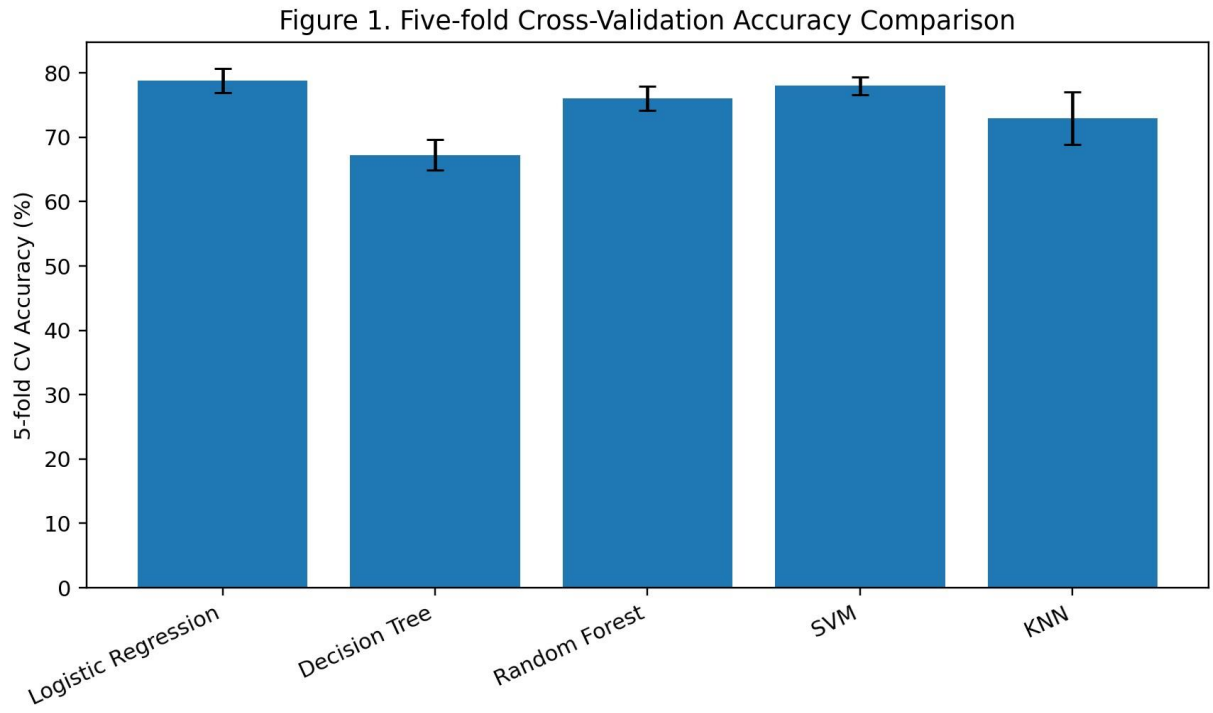


Figure 1. Five-fold cross-validation accuracy; error bars represent fold-to-fold standard deviation.

6.2 Primary 80:20 Held-Out Test Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	70.78%	60.00%	50.00%	54.55%	81.30%
Decision Tree	68.18%	55.32%	48.15%	51.49%	63.57%
Random Forest	74.03%	65.91%	53.70%	59.18%	81.54%
SVM	74.03%	65.22%	55.56%	60.00%	79.64%
KNN	75.32%	66.00%	61.11%	63.46%	78.99%

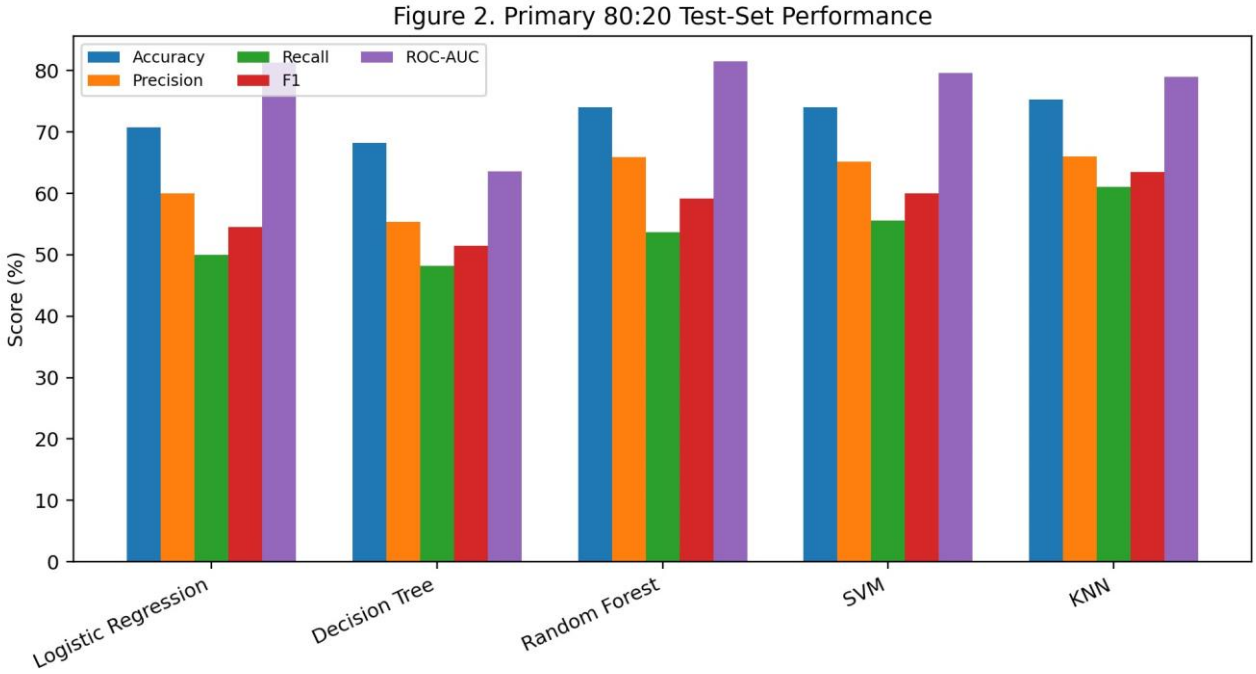


Figure 2. Performance of the five classifiers on the primary held-out 20% test set.

6.3 Repeated 80:20 Robustness Analysis

To examine sensitivity to the random partition, the fixed tuned configurations were evaluated over five additional stratified 80:20 splits (seeds 11, 22, 33, 44, and 55). The values below are descriptive robustness statistics and are distinct from the primary test split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	76.62 ± 1.59	73.22 ± 6.35	53.70 ± 5.40	61.62 ± 2.78	84.84 ± 2.09
Decision Tree	70.65 ± 1.48	58.10 ± 2.16	58.52 ± 4.83	58.24 ± 2.94	67.86 ± 2.00
Random Forest	75.97 ± 4.33	68.73 ± 7.25	58.89 ± 8.43	63.12 ± 6.43	84.36 ± 2.93
SVM	76.10 ± 2.81	70.99 ± 6.11	54.81 ± 8.45	61.45 ± 5.41	84.06 ± 2.97
KNN	74.55 ± 3.48	65.43 ± 4.75	57.78 ± 7.10	61.31 ± 5.89	79.64 ± 3.11

6.4 Random Forest Hyperparameter Search

The best five-fold cross-validated F1-score during the randomized search was 65.11%. The selected configuration is reported in the methodology so that the experiment can be reproduced.

6.5 Feature Importance

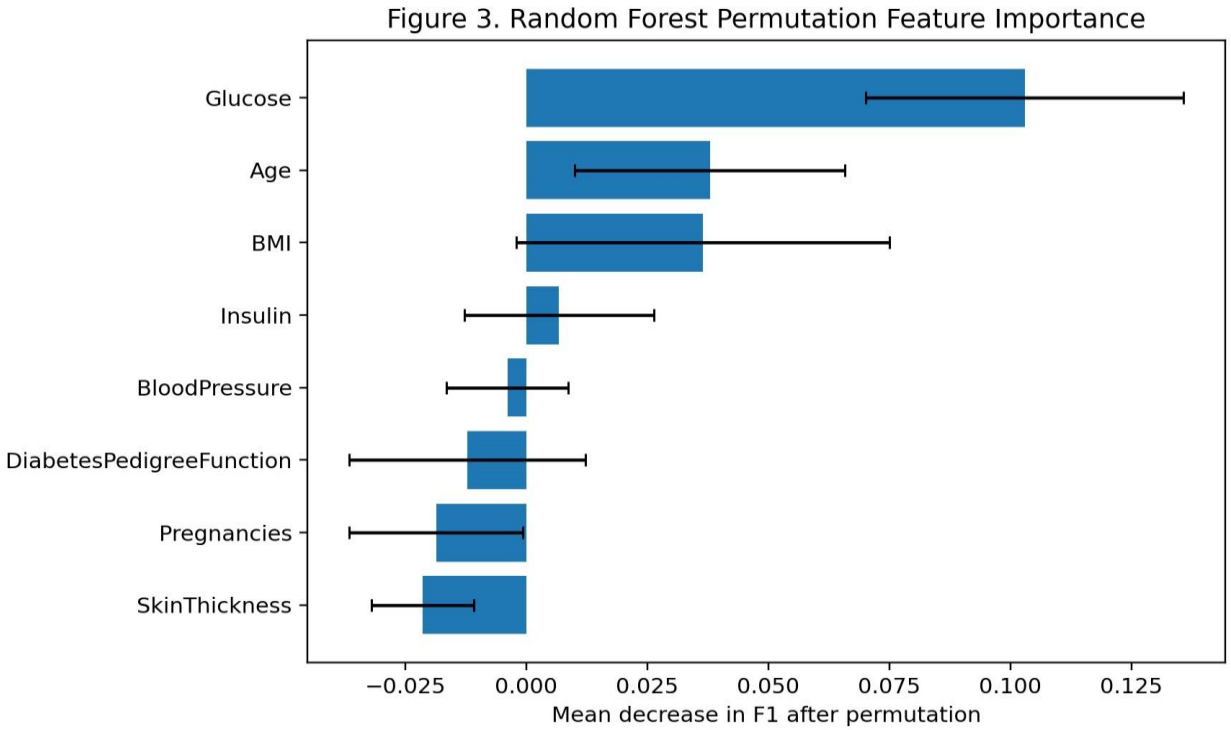


Figure 3. Permutation importance of Random Forest features on the primary held-out test set.

Glucose showed the largest positive mean permutation importance (0.103), followed by Age (0.038) and BMI (0.036). Some variables produced near-zero or negative mean changes in F1 under this particular test set. Negative permutation importance does not mean that a feature is medically harmful; it indicates that shuffling that feature did not improve the measured F1 score in this experiment.

6.6 Confusion Matrix and Decision-Tree Learning Curve

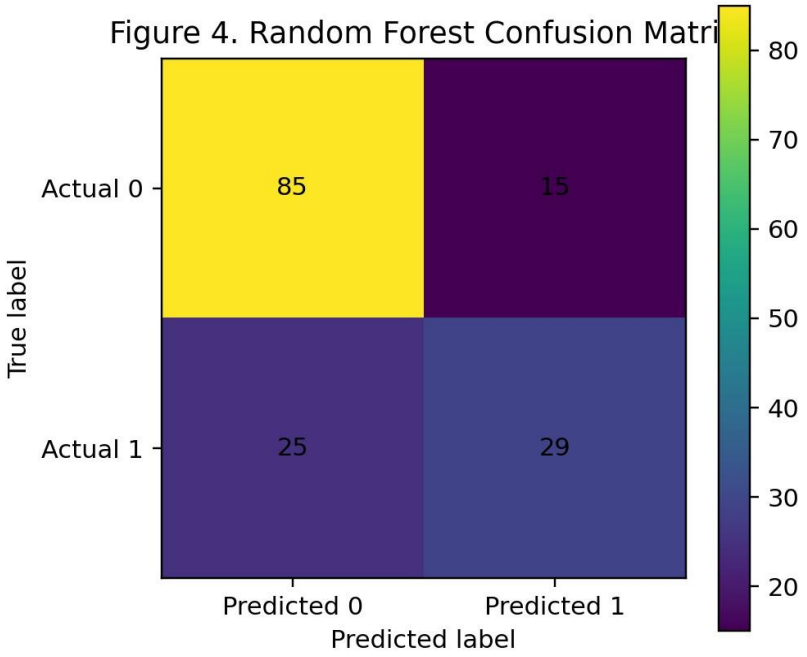


Figure 4. Tuned Random Forest confusion matrix. The matrix contains TN=85, FP=15, FN=25, and TP=29 on the primary test set.

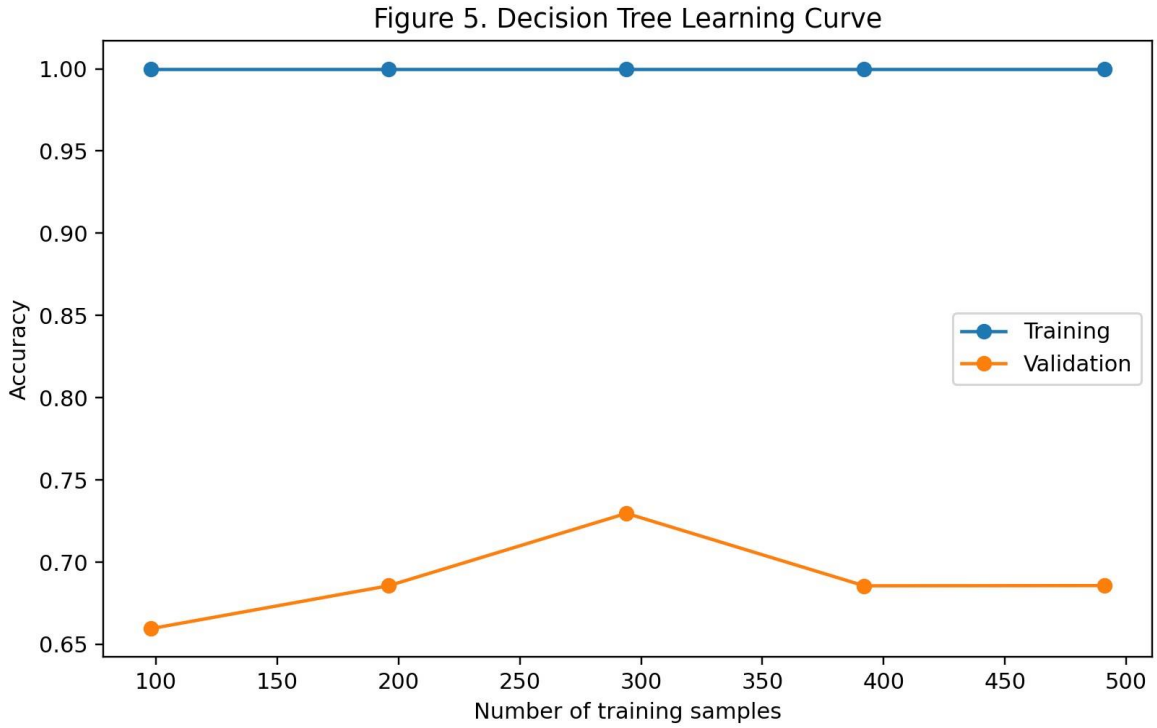


Figure 5. Decision Tree learning curve using five-fold cross-validation.

7. Discussion

The cross-validation results show that Logistic Regression and SVM provided the strongest average accuracy among the evaluated baseline models, while Random Forest remained competitive. On the primary held-out split, the tuned Random Forest achieved 74.03% accuracy, 65.91% precision, 53.70% recall, 59.18% F1, and 81.54% ROC-AUC.

These values should be interpreted as results for the supplied dataset and experimental configuration, not as clinical performance estimates.

The Decision Tree had the lowest cross-validated accuracy and ROC-AUC among the five models. A likely methodological explanation is that a single tree can form highly specific decision boundaries and may overfit local patterns, particularly in a relatively small dataset. The learning curve is therefore useful for diagnosing the relationship between training and validation behaviour. Ensemble averaging in Random Forest reduces dependence on a single tree and can provide more stable nonlinear decision boundaries.

The feature-importance analysis indicates that glucose was the most influential variable under the chosen permutation/F1 procedure. Age and BMI also showed positive mean importance. Because permutation importance can be affected by correlated predictors and by the particular held-out sample, these rankings should not be interpreted as causal medical findings.

8. Tools and Implementation

The experimental workflow was implemented in Python using Pandas and NumPy for data handling and scikit-learn for preprocessing, model training, cross-validation, hyperparameter search, evaluation, and permutation importance.Matplotlib was used for the figures. The original project was developed in Google Colab.

9. Limitations

- The Pima dataset is relatively small and represents a specific population; generalization to other populations requires external validation.
- The five additional 80:20 runs provide robustness information but do not replace prospective clinical validation.
- Random Forest tuning used a limited randomized search budget of 10 configurations; broader optimization may produce different settings.
- Permutation importance is model- and sample-dependent and does not establish causal relationships.
- The model is a research prototype and should not be used as a standalone diagnostic system.

10. Future Scope

Future work should evaluate repeated or nested cross-validation, larger and more diverse datasets, systematic hyperparameter optimization, calibration analysis, explainable-AI methods such as SHAP, external validation, and prospective clinical evaluation. Threshold selection may also be optimized according to whether the application prioritizes sensitivity or specificity.

11. Conclusion

The updated study carries out a more rigorous comparison of five machine-learning classifiers when it comes to predicting diabetes in its early stages. Through the use of a stratified 80:20 held-out data split together with five-fold cross-validation, leakage-aware preprocessing, tuning of the Random Forest hyperparameters, repeated split analysis, examination of feature importance, analysis of confusion matrices, and the use of learning curves, the evaluation overcomes a number of the shortcomings associated with a single-model, single-split experiment. The findings confirm the value of machine-learning approaches for research into diabetes-risk prediction, but at the same time stress that limitations relating to the dataset and the lack of clinical validation mean that the results cannot be regarded as suitable for use as a clinical diagnostic tool.

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